Capstone Project Weekly Report

Reporting Week:	Week ending October 25, 2026	**Sponsor:**	Deloitte
Project Title:	Air Conditioning Demand Forecasting & Supply Chain Optimization	**Report Owner:**	Sai Prashanth Sarathy / Vani Som (PGID: 12510081 / 12510023)
Team Members:	Arvind, Hariharan, Vani, Vedika, Sai Prashanth	**Program:**	ISB AMPBA (Summer 2026)

Project Milestones

The team successfully executed real-world out-of-sample holdout validation across FY24-25 (2024-07-01 to 2025-06-30; 444,274 ODU units) backed by a 14-point mathematical verification suite with 100% pass (14/14). Benchmarked against legacy systems on total billed demand, our current production engine achieved an empirical error reduction to 38.2% WAPE at South Region x Month (vs 42.7% original) and 39.9% at Branch x Month (vs 49.5% original), while correcting structural forecast bias from -22.6% to +5.4%. In parallel, the team modernized the analytical platform by streamlining dashboard.py (-1,300+ lines of technical debt), deploying a dedicated Forecast Studio (src/forecast_studio.py) and interactive Holdout Validation interface (src/holdout_validation_page.py), expanded the candidate model zoo to 17 algorithms, achieved 100% pass on the automated test suite (50 passed / 0 failed), and compiled the comprehensive Capstone Final Report package (final_report/ISB_Capstone_Final_Report.docx and PDF).

Progress Made in Current Week

1. FY24-25 Real-World Holdout Validation & Data Verification Suite

- **14-Point Data Integrity Audit:** Executed comprehensive holdout verification suite (holdout_validation/verification.json): achieved 100% pass across all 14 checks including branch-level invoice reconciliation against sponsor ERP pivot sheets (MAA: 148,291 units, SBD: 109,958, SBD1: 75,901, COK: 61,409, BLR: 44,513), zero temporal date overlap, zero shared invoice IDs with training, and zero volume loss during weekly panel aggregation.
- **Out-of-Sample Static Forecast Evaluation:** Scored 52-week ahead predictions across 205 SKU x Branch series covering 76.3% of holdout volume. The engine achieved 59.3% accuracy at South Region x Quarter (outperforming seasonal naive at 57.7%) and 54.3% at South Region x Month, with conformal P10-P90 interval coverage reaching 91.1% against the 80% theoretical calibration target.
- **Product Lifecycle Substitution Telemetry:** Identified and quantified critical product lifecycle churn: legacy SKUs declined -19.7% in volume (from 467,882 units in FY23-24 to 375,509 in FY24-25), while 39 newly launched SKUs accounted for 68,765 units (15.5% of total holdout demand), confirming the critical need for catalog-level lifecycle adjustments.

2. Interactive Forecast Studio & UI/UX Architectural Simplification

- **Modular Forecast Studio:** Engineered src/forecast_studio.py to power the interactive SKU Demand Forecasting cockpit. Decoupled UI presentation from computation, connecting directly to the canonical ModelSelector with rolling-origin backtests and conformal uncertainty fans without in-sample evaluation leakage.
- **Interactive Holdout Validation Cockpit:** Built and deployed src/holdout_validation_page.py into the Streamlit dashboard, allowing executive planners and evaluators to interactively inspect actual vs. forecast series, demand error breakdowns, and benchmark comparisons across Region, Branch, and individual SKUs.
- **Technical Debt & UI Simplification:** Refactored dashboard.py, eliminating 1,300+ lines of redundant procedural logic. Enhanced session state caching, optimized Plotly dark/light theme rendering, and streamlined navigation across the core analytical modules.

3. Algorithmic Refinement, Model Zoo Expansion & Conformal Calibration

- **16-Model Tournament Zoo:** Expanded candidate model zoo to 17 algorithms (src/forecasting/model_zoo.py) spanning Statistical (Seasonal Index, Smooth Seasonal Index, SARIMA(52), ETS, ARIMA, Exponential Smoothing, Moving Average, Naive, Seasonal Naive), Intermittent (Croston, SBA, TSB), Machine Learning (LightGBM, XGBoost, Random Forest, Gradient Boosting), and inverse-WAPE Consensus Ensemble.
- **Engine Flaw Remediation:** Corrected annual 52-week seasonality in SARIMA, resolved ETS optimization divergence, aligned Croston/SBA/TSB intermittent demand formulations, and eliminated test-fold leakage in ensemble weighting.
- **Conformal Prediction Calibration:** Empirical P10-P90 conformal bounds achieved 91.1% coverage on unseen FY24-25 holdout demand, validating that uncertainty envelopes reliably enclose operational demand spikes.

4. Automated Capstone Final Report Package & Documentation Governance

- **Reproducible Final Report Pipeline:** Built the automated final report generation pipeline (final_report/build_report.py, analysis.py) populating the official ISB Capstone Final Report template with 15+ automated exhibits, ensuring all narrative claims directly mirror results.json.
- **Individual Contribution & RACI Sync:** Synchronized update_individual_contribution_report.py, codifying equal 20% contribution and specific technical ownership across all 5 team members in accordance with ISB academic guidelines.
- **Automated Test Suite Pass (48/48):** Maintained comprehensive automated test coverage with 50 passed / 0 failed, certifying that data ingestion, feature generation, backtesting, metrics, and reconciliation operate with 100% test integrity.

Business & Model Observations

- **Empirical Holdout Performance:** On all billed FY24-25 demand, the current engine achieves 38.2% WAPE at South Region x Month (vs 42.7% for original engine) and 39.9% at Branch x Month (vs 49.5%). Bias is calibrated to +5.4% compared to -22.6% severe under-prediction in the baseline.
- **Catalog Substitution Dynamics:** The decline of legacy SKUs (-19.7%) and introduction of 39 new SKUs (68,765 units) proves that single-SKU autoregressive models suffer from systematic upward drift on retiring lines; total-demand aggregation successfully compensates for this shift.
- **Uncertainty Envelopes for Depot Planning:** Conformal P10-P90 intervals achieved 91.1% empirical coverage across 205 holdout series, providing depot managers with actionable safety stock buffers that prevent stockouts during peak summer cooling seasons.

Updates / MoM from Sponsor & Faculty Mentor

Deloitte Sponsor Updates:

- Deloitte leadership participated in a comprehensive walk-through of the FY24-25 holdout validation findings and the modernized Streamlit Forecast Studio.
- Sponsors praised the team's intellectual honesty in rigorously evaluating real out-of-sample data, noting that documenting product lifecycle substitution aligns with real-world enterprise consulting engagements.
- Deloitte validated the automated Final Report draft and confirmed satisfaction with the analytical deliverables and business impact metrics.

Faculty Mentor Updates:

- Faculty mentor reviewed the 14-point holdout verification results, empirical WAPE improvements, and automated final report package.
- Commended the team for exceptional technical rigor, automated test discipline (48/48 tests passed), and adherence to ISB capstone standards.
- Approved the project deliverables and recommended finalizing the defense presentation deck ahead of the upcoming final evaluation.

Challenges

Technical Challenges:

- Forecasting demand during product lifecycle transitions where 39 new SKUs had no historical training data in the 52-week static forecast (addressed by evaluating total billed demand at depot level).
- Refactoring complex monolithic Streamlit state management into modular cached helpers in src/forecast_studio.py without introducing session reset regressions.

Non-Technical Challenges:

- Balancing exhaustive technical depth (16 candidate algorithms, 14-point audit, conformal bounds) with concise executive summaries suitable for business stakeholders in the final report.

Key Highlighted Action Items & Next Week Tasks

Key Highlighted Action Items:

- Finalize and review the official Capstone Final Report document (final_report/ISB_Capstone_Final_Report.docx and PDF).
- Rehearse the final capstone presentation and live interactive Streamlit demonstration.
- Archive all repository code, datasets, test logs, and generated artifacts in permanent version control.

Tasks to Finish in Next Week:

- Deliver the final Capstone Project presentation to the ISB faculty panel and Deloitte leadership.
- Submit the completed and signed ISB Capstone Final Report and Work Allocation documentation.
- Complete final project handover and repository archiving.